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NIMCET 2008

1. The value of λ for which the volume of parallelepiped formed by the vectors $\hat{\imath}+\lambda\hat{\jmath}+\hat{k}, \hat{\jmath}+\lambda\hat{k}$ and $\lambda\hat{\imath}+\hat{k}$ is minimum is given by

(a)-3 (b)3 (c)$\frac{1}{\sqrt{3}}$ (d)$-\sqrt{3}$

2. The value of λ such that the four points whose positions vectors are $3\hat{\imath}-2\hat{\jmath}+\lambda\hat{k}, 6\hat{\imath}+3\hat{\jmath}+\hat{k}, 5\hat{\imath}+7\hat{\jmath}+3\hat{k}$ and $2\hat{\imath}+2\hat{\jmath}+6\hat{k}$ are coplanar is

(a)-6 (b)4 (c)5 (d)8

3. Let $A=2\hat{\imath}+\hat{\jmath}-2\hat{k}$ and $B=\hat{\imath}+\hat{\jmath}$. If C is a vector such that $A\cdot C=|C|, |C-A|=2\sqrt{2}$ and the angle between $A\times B$ and C is 30°, then $|(A\times B)\times C|$ is equal to

(a)$\frac{2}{3}$ (b)$\frac{3}{2}$ (c)2 (d)3

4. A rigid body is rotating at the rate of 3 rads about an axis AB, where A and B are the points $(1,-2,1)$ and $(3,-4,2)$. The velocity of the point P at $(5,-1,-1)$ of the body is

(a)$3\hat{\imath}+8\hat{\jmath}+10\hat{k}$ (b)$\frac{3\hat{\imath}+8\hat{\jmath}+10\hat{k}}{3}$

(c)$\frac{2\hat{\imath}-2\hat{\jmath}+\hat{k}}{3}$ (d)$4\hat{\imath}+\hat{\jmath}+2\hat{k}$

5. If $A+B+C=0. |A|=3, |B|=5, |C|=7$ then the angle between A and B is

(a)$\frac{\pi}{6}$ (b)$\frac{2\pi}{3}$ (c)$\frac{5\pi}{3}$ (d)$\frac{\pi}{4}$

NIMCET 2009

6. The vector $B=3\hat{\imath}+4\hat{k}$ is to be written as the sum of a vector B_1 parallel to $A=i+j$ and a vector B_2 perpendicular to A, then B_1 is

(a)$\frac{3}{2}(\hat{\imath}+\hat{\jmath})$ (b)$\frac{2}{3}(\hat{\imath}+\hat{\jmath})$

(c)$\frac{1}{2}(\hat{\imath}+\hat{\jmath})$ (d)None of these

7. If a, b and c are unit vectors, then $|a-b|^2+|b-c|^2+|c-a|^2$ does not exceed

(a)9 (b)4 (c)8 (d)6

8. a, b, c are non-coplanar unit vectors such that $a\times(b\times c)=\frac{b+c}{\sqrt{2}}$, then the angle between a and b is

(a)$\frac{\pi}{4}$ (b)$\frac{3\pi}{4}$ (c)$\frac{\pi}{2}$ (d)π

NIMCET 2010

9. If C is the middle point AB and P is any point outside AB, then

(a)$PA+PB=PC$ (b)$PA-PB=2PC$

(c)$PA+PB+PC=0$ (d)$PA+PB+2PC=0$

10. Let a, b and c be three non zero vectors no two of which are collinear and the vector $a+b$ is collinear with c while $b+c$ is collinear with a then $a+b+c$ is equal to

(a)a (b)b (c)c (d)none

NIMCET 2011

11. If θ is the angle between a and b and $|a\times b|=|a\cdot b|$, then θ is equal to

(a)0 (b)π (c)$\pi/2$ (d)$\pi/4$

12. If a, b and c are unit coplanar vectors, then the scalar triple product $[2a-b, 2b-c, 2c-a]$ will be equal to

(a)0 (b)1 (c)$-\sqrt{3}$ (d)$\sqrt{3}$

13. Let $a=x\hat{\imath}-3\hat{\jmath}-\hat{k}$ and $b=2x\hat{\imath}+x\hat{\jmath}-\hat{k}$. Suppose that the angle between a and b is acute and the angle between, b and the positive direction of the y-axis lies between $\frac{\pi}{2}$ *and* π. Then, the set of all possible values of x is

(a)$\{1, 2\}$ (b)$\{-2,-3\}$

(c)$\{x: x<0\}$ (d)$\{x: x>0\}$

14. Let $v=2\hat{\imath}+\hat{\jmath}-\hat{k}$ and $w=\hat{\imath}+3\hat{k}$. If u is a unit vector, then the maximum value of the scalar triple product [u v w]

(a)-1 (b)$-\sqrt{10}-\sqrt{6}$

(c)$\sqrt{59}$ (d)$\sqrt{10}+\sqrt{6}$

15. ABCD is a parallelogram with AC and BD as diagonals. Then $AC-BD$ is equal to

(a)4 AB (b)3 AB (c)2 AB (d)AB

NIMCET 2012

16. The equation of the plane passing through the point (1, 2, 3) and having the vector $N=3\hat{\imath}-\hat{\jmath}+2\hat{k}$ as its normal is

(a)$2x-y+3z+7=0$ (b)$3x-y+2z+7=0$

(c)$3x-y+2z=7$ (d)$3x+y+2z=7$

17. If a, b and c are unit vectors such that $a+b+c=0$, then the value of $a\cdot b+b\cdot c+c\cdot a$ is

(a)2/3 (b)$-2/3$ (c)3/2 (d)$-3/2$

18. If the vectors $a=(1,x,-2)$ *and* $b=(x,3,-4)$ are mutually perpendicular, then the value of x is

(a)-2 (b)2 (c)4 (d)-4

19. If a, b, c are non-coplanar vectors and λ is a real number, then the vectors $a+3b+3c, \lambda b+4c$ and $(2\lambda-1)c$ are non-coplanar for

(a)all values of λ

(b)all except one value of λ

(c)all except two values of λ

(d)no value of λ

20. If $a+b+c=0, |a|=3, |b|=5, |c|=7$ are non-coplanar vectors and λ is a real number, then angle between the vectors a and b is

(a)$\frac{\pi}{2}$ (b)$\frac{\pi}{3}$ (c)$\frac{\pi}{4}$ (d)$\frac{\pi}{6}$

21. If $\theta(0\le\theta\le\pi)$ is the angle between the vectors a and b, then $\frac{|a\times b|}{a\cdot b}$ is equal to

(a)$-\cot\theta$ (b)$\tan\theta$ (c)$-\tan\theta$ (d)$\cot\theta$

NIMCET 2013

22. If $a=\hat{\imath}-\hat{k}$ *and* $c=\hat{\imath}+\hat{\jmath}-\hat{k}$ *and* $3\hat{\imath}-\hat{k}$ from three conterminous edges of a parallelopiped , then the volume of parallelopiped is

(a)$-\hat{\imath}+\hat{\jmath}+2\hat{k}$ (b)$2\hat{\imath}-\hat{\jmath}+2\hat{k}$

(c)$\hat{\imath}-\hat{\jmath}-2\hat{k}$ (d)$\hat{\imath}+\hat{\jmath}-2\hat{k}$

23. If $a=\hat{\jmath}-\hat{k}$ and $c=\hat{\imath}-\hat{\jmath}-\hat{k}$, then vector b satisfying $(a\times b)+c=0$ and $a\cdot b=3$ is

(a)$-\hat{\imath}+\hat{\jmath}-2\hat{k}$ (b)$2\hat{\imath}-\hat{\jmath}+2\hat{k}$

(c)$\hat{\imath}-\hat{\jmath}-2\hat{k}$ (d)$\hat{\imath}+\hat{\jmath}-2\hat{k}$

24. Force $3\hat{\imath}+2\hat{\jmath}+5\hat{k}$ and $2\hat{\imath}+\hat{\jmath}-3\hat{k}$ are acting on a particle and displace it from the point $2\hat{\imath}-\hat{\jmath}-3\hat{k}$ to the point $4\hat{\imath}-2\hat{\jmath}+7\hat{k}$, then the work done by the force is
(a)18 units (b)30 units
(c)27 units (d)36 units

25. If the vectors $2\hat{\imath}-3\hat{\jmath}, \hat{\imath}+\hat{\jmath}-\hat{k}$ and $3\hat{\imath}-\hat{k}$ from three conterminous edges of a parallelopiped, then the volume of parallelopiped
(a)8 (b)10 (c)4 (d)14

26. The area of the parallelogram whose diagonals are $a=3\hat{\imath}+\hat{\jmath}-2\hat{k}$ and $b=\hat{\imath}-3\hat{\jmath}+4\hat{k}$ is
(a)$10\sqrt{3}$ (b)$5\sqrt{3}$
(c)$10\sqrt{2}$ (d)$5\sqrt{2}$

27. If the three vectors $2\hat{\imath}-\hat{\jmath}+\hat{k}, \hat{\imath}+2\hat{\jmath}-3\hat{k}$ and $3\hat{\imath}+\lambda\hat{\jmath}+5\hat{k}$ are coplanar, then λ is
(a)-1 (b)-2 (c)-3 (d)-4

28. If a, b and c are the position vectors of three vertices A, B and C of a triangle, respectively. Then, the area of this triangle is given by
(a)$\frac{1}{2}(a\times b)$ (b)$\frac{1}{2}|a\times b+b\times c+c\times a|$
(c)$a\times b+b\times c+c\times a$ (d)none

NIMCET 2014

29. Constant forces $P=2\hat{\imath}-5\hat{\jmath}+6\hat{k}$ and $Q=-\hat{\imath}+2\hat{\jmath}-\hat{k}$ act on a particle. The work done when the particle is displaced from A whose position vector is $4\hat{\imath}-3\hat{\jmath}-2\hat{k}$ to B whose position vector $6\hat{\imath}+\hat{\jmath}-3\hat{k}$, is
(a)10 units (b)-15 units
(c)-50 units (d)25 units

30. For the vector $a=-4\hat{\imath}+2\hat{\jmath}, b=2\hat{\imath}+\hat{\jmath}$ and $c=2\hat{\imath}+3\hat{\jmath}$, if $c=ma+nb$, then the value of $m+n$ is
(a)$\frac{1}{2}$ (b)$\frac{3}{2}$ (c)$\frac{5}{2}$ (d)$\frac{7}{2}$

31. $A=4\hat{\imath}+3\hat{\jmath}+\hat{k}, B=2\hat{\imath}-\hat{\jmath}+2\hat{k}$, then the unit vectors N perpendicular to vector A and B such that A, B, N from a right handed system is
(a)$\frac{1}{\sqrt{185}}[7\hat{\imath}-6\hat{\jmath}-10\hat{k}]$ (b)$\frac{1}{7}[6\hat{\imath}+2\hat{\jmath}+3\hat{k}]$
(c)$\frac{1}{\sqrt{21}}[2\hat{\imath}+4\hat{\jmath}-\hat{k}]$ (d)$\frac{1}{\sqrt{21}}[-2\hat{\imath}-4\hat{\jmath}+\hat{k}]$

32. The sum of two vectors a and b is a vector c such that $|a|=|b|=|c|=2$. Then, the magnitude of $a-b$ is equal to
(a)$2\sqrt{3}$ (b)2 (c)$\sqrt{3}$ (d)0

33. If a vector a makes an equal angle with the coordinate axes and has magnitude 3, then the angle between a and each of the three coordinates axes is
(a)$\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b)$\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(c)$\frac{\pi}{6}$ (d)$\frac{\pi}{3}$

34. The median AD of ΔABC is bisected at E and BE is extended to meet the side AC in F. The AF: AC=
(a)1:3 (b)2:1 (c)1:2 (d)3:1

NIMCET 2015

35. If a and b are vectors such that $|a|=13, |b|=5$ and $a\cdot b=60$, then the value of $|a\times b|$ is
(a)625 (b)225 (c)45 (d)25

36. If $a=4\hat{\imath}+6\hat{\jmath}$ and $b=3\hat{\imath}+4\hat{k}$, then the vector form of the component of a along b is
(a)$\frac{18}{10\sqrt{3}}(3\hat{\imath}+4\hat{k})$ (b)$\frac{18}{5}(3\hat{\imath}+4\hat{k})$
(c)$\frac{18}{\sqrt{3}}(3\hat{\imath}+4\hat{k})$ (d)$(3\hat{\imath}+4\hat{k})$

37. If a and b are vector in space, given by $a=\frac{\hat{\imath}-2\hat{\jmath}}{\sqrt{5}}$ and $b=\frac{2\hat{\imath}+\hat{\jmath}+3\hat{k}}{\sqrt{14}}$, then the value of $(2a+b).[(a\times b)\times(a-2b)] is$
(a)3 (b)4 (c)5 (d)6

38. If $a=\hat{\imath}-\hat{k}, b=x\hat{\imath}+\hat{\jmath}+(1-x)\hat{k}$ and $c=y\hat{\imath}+x\hat{\jmath}+(1+x-y)\hat{k}$, then [a b c] depends on
(a)neither x nor y (b)Only x
(c)Only y (d)Both x and y

39. Let a and b be two vectors. Which of the following vectors are not perpendicular to each other?
(a)$(a\times b)$ and a (b)$(a+b)$ and $a\times b$
(c)$(a+b)$ and $a-b$ (d)$a-b$ and $a\times b$

40. Let $a=\hat{\imath}+\hat{\jmath}+\hat{k}, b=\hat{\imath}-\hat{\jmath}+\hat{k}$ and $c=\hat{\imath}-\hat{\jmath}-\hat{k}$ be three vector v in the plan of a and b whose projection on $\frac{c}{|c|}$ is $\frac{1}{\sqrt{3}}$, is
(a)$3\hat{\imath}-\hat{\jmath}+3\hat{k}$ (b)$\hat{\imath}-3\hat{\jmath}+3\hat{k}$
(c)$5\hat{\imath}-2\hat{\jmath}+5\hat{k}$ (d)$2\hat{\imath}-\hat{\jmath}+3\hat{k}$

41. If a, b and c are the position vectors of the vertices A, B, C of a triangle ABC, then the area of the triangle ABC is
(a)$\frac{1}{2}|a\times b+b\times c+c\times a|$
(b)$|a\times b|$
(c)$\frac{1}{2}|a\times b-b\times c-c\times a|$
(d)$|a\times(b\times c)|$

42. If $AC=2\hat{\imath}+\hat{\jmath}+\hat{k}$ and $BD=-\hat{\imath}+3\hat{\jmath}+2\hat{k}$, then the area of the quadrilateral ABCD is
(a)$\frac{5}{2}\sqrt{3}$ (b)$5\sqrt{3}$ (c)$\frac{15}{2}\sqrt{3}$ (d)$10\sqrt{3}$

NIMCET 2016

43. If C is the mid-point of AB and P is any point outside AB, then
(a)$PA+PB=2PC$ (b)$PA+PB=PC$
(c)$PA+PB+2PC=0$ (d)$PA+PB+PC=0$

44. The volume of the parallelopiped determined $U=\hat{\imath}+2\hat{\jmath}-\hat{k}, V=-2\hat{\imath}+3\hat{k}$ and $W=7\hat{\jmath}-4\hat{k}$ is
(a)21 (b)22
(c)23 (d)24

45. The vector perpendicular to the plane passing through$(1,-1,0), (-2,1,-1)$ and $(-1,1,2)$ is
(a)$6\hat{\imath}+6\hat{\jmath}$ (b) $6\hat{\imath}+7\hat{k}$
(c) $7\hat{\imath}+6\hat{k}$ (d) $7\hat{\imath}+8\hat{k}$

46. If a, b and a+b are vectors of magnitude a, then the magnitude of vector a-b is

(a)$\sqrt{2}a$ (b) $\sqrt{3}a$
(c)$2a$ (d) $3a$

47. Let a, b and b be three non- zero vectors, no two of which are collinear. If the vector $a+2b$ is collinear with c and $b+3c$ is collinear with a, then $a+2b+6c$ is equal to
(a)λa (b) λb (c) λc (d)0

48. If a, b are vectors such that $|a+b|=\sqrt{29}$ *and* $a\times(2\hat{\imath}+3\hat{\jmath}+4\hat{k})=(2\hat{\imath}+3\hat{\jmath}+4\hat{k})\times b$, then a possible value of$(a+b).(-7\hat{\imath}+2\hat{\jmath}+3\hat{k})$ is
(a)0 (b)3 (c)4 (d)8

NIMCET 2017

49. The direction cosines of the vector $a=(-2\hat{\imath}+\hat{\jmath}-5\hat{k})$ are
(a)$-2,1,-5$ (b)$\frac{1}{3},-\frac{1}{6},-\frac{5}{6}$
(c)$\frac{2}{\sqrt{30}},\frac{1}{\sqrt{30}},\frac{5}{\sqrt{30}}$ (d) $\frac{-2}{\sqrt{30}},\frac{1}{\sqrt{30}},\frac{-5}{\sqrt{30}}$

50. If $\bar{a},\bar{b},\bar{c}$ are vectors such that $\vec{a}+\vec{b}+\vec{c}=0$ *and* $|\vec{a}|=7,|\vec{b}|=5,|\vec{c}|=3$,then the angle between the vectors $\vec{b}$ *and* $\vec{c}$ *is*
(a)60° (b)30° (c)45° (d)90°

51. If $a\hat{\imath}+\hat{\jmath}+k,\hat{\imath}+b\hat{\jmath}+k,i+\hat{\jmath}+c\hat{k}\ (a\neq b\neq c\neq 1)$ Are coplanar, then the value of $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}$ is
(a)-1 (b)$-\frac{1}{2}$ (c)$\frac{1}{2}$ (d)1

52. Let $\vec{a},\vec{b}$*and* $\vec{c}$ be three vector having magnitudes 1,1 and 2, respectively. If $\vec{a}\times(\vec{a}\times\vec{c})-\vec{b}=0$, then the acute angle between $\vec{a}$ *and* $\vec{c}$ is
(a)$\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{3}$ (d)None

53. Let $\vec{a},\vec{b},\vec{c}$ be vector such that $|a|=2,|\vec{b}|=3,|c|=5$ *and*$\vec{a}+\vec{b}+\vec{c}=\vec{0}$. The value of $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ is
(a)38 (b)-38 (c)19 (d)-19

54. If $\vec{a}=(\hat{\imath}+2\hat{\jmath}-3\hat{k})$*and*$\vec{b}=(3\hat{\imath}-\hat{\jmath}+2\hat{k})$ then the angle between $(a+b)$ *and* $(a-b)$ *is*
(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{2\pi}{3}$

NIMCET 2018

55. If a, b and c are unit vectors, then $|a-b|^2+|b-c|^2+|c-a|^2$ does not exceed
(a)4 (b)9 (c)8 (d)6

56. The vector $\vec{a}=\alpha\hat{\imath}+2\hat{\jmath}+\beta\hat{k}$ lies in the plane of the vectors $\vec{b}=\hat{\imath}+\hat{\jmath}$ *and* $\vec{c}=\hat{\jmath}+\hat{k}$ and bisects the angle between $\vec{b}$ *and*$\vec{c}$. Then, which one of the following gives possible values of α *and*β?
(a)$\alpha=2,\beta=2$ (b) $\alpha=1,\beta=2$
(c) $\alpha=2,\beta=1$ (d) $\alpha=1,\beta=1$

57. Forces $4\hat{\imath}-3\hat{\jmath}+7\hat{k}$ *and* $-2\hat{\imath}+2\hat{\jmath}-8\hat{k}$ are acting on a particle and displace it form the point (5,7,1) to (2,5,-6), then the work done by the force is
(a)25 (b)9 (c)15 (d)7

58. A bird is flying in a straight line with velocity vector $10\hat{\imath}+6\hat{\jmath}+k$, measured in km/h. If starting points is (1,2,3), how much time does it take to reach a point in space that is 13m high from ground?
(a)600 seconds (b)360 seconds
(c)36 seconds (d)60 seconds

NIMCET 2019

59. A particle P starts form the point $z_0=1+2i$*, where* $i=\sqrt{-1}$, it moves first horizontally away from the origin by 5 units and then vertically away from origin by 3 units to reach a point z_1.
From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{\imath}+\hat{\jmath}$ and, then it moves through an angle $\pi/2$ in an anticlockwise direction on a circle with centre at origin, to reach a point z_2. The point z_2 is given by
(a)$6+7\hat{\imath}$ (b)$-7+6\hat{\imath}$
(c)$7+6\hat{\imath}$ (d)$-6+7\hat{\imath}$

60. $\vec{a}$ *and*$\vec{b}$ are non- zero non-collinear vectors such that $|\vec{a}|=2,\vec{a}\,.\,\vec{b}=1$ and the angle between $\vec{a}$ *and* $\vec{b}$ is $\frac{\pi}{3}$.
If $\vec{r}$ any vector satisfying $\vec{r}.\vec{a}=2,\ \vec{r}.\vec{b}=8,(\vec{r}+2\vec{a}-10\vec{b}).(\vec{a}\times\vec{b})=6\vec{r}+2\vec{a}-10\vec{b}=\lambda(\vec{a}\times\vec{b})$*, then*$\lambda=$
(a)$\frac{1}{2}$ (b)2 (c)$\frac{4}{\sqrt{3}}$ (d)3

61. The position vectors of the vertices A,B,C of a tetrahedron ABCD are $\hat{\imath}+\hat{\jmath}+\hat{k},\hat{\imath}$ and $3\hat{\imath}$ respectively and the altitude for the vertex D to the opposite face ABC meets the face at E. If the length of ED is 4 and the volume of tetrahedron
is $\frac{2\sqrt{2}}{3}$, then length of DE is
(a)1 (b)2 (c)3 (d)4

62. In a parallelogram ABCD, P is the mid-point of AD. Also, BP and AC intersect at Q. then, AQ: QC=
(a)1:3 (b)3:1 (c)2:1 (d)1:2

63. The median AD of ΔABC is bisected at E and BE is extended to meet the side AC in F. The AF: AC=
(a)1:3 (b)2:1 (c)1:2 (d)3:1

64. The value of non-zero scalars α *and* β that for all vectors a *and* b, such that $\alpha(a+2b)-\beta(4b-a)=0$ is
(a)$\alpha=2,\beta=1$ (b) $\alpha=-2,\beta=-3$
(c) $\alpha=1,\beta=3$ (d) $\alpha=-2,\beta=3$

65. A force of 78 grams acts at the point(2,3,5). The direction ratios of the line of action being (2,2,1). The magnitude of its moments of the line joining The origin to the point(12,3,4) is
(a)24 (b)136 (c)36 (d)0

NIMCET 2020

66. If a, b, c are three non-coplanar vectors, then(a+b+c).[(a+b)×(a+c)]=
(a)0 (b)[a b c] (c)2[a b c] (d)- [a b c]

67. Two forces F1 and F2 are used to pull a car, which met an accident. The angle between the two forces is θ. Find

the value of θ for which the resultant force is equal to$\sqrt{\{F_1^2 + F_2^2\}}$
(a) θ=0° (b) θ=45°
(c) θ=90° (d) θ=135°

68. If a, b, c, d are four vectors such that a+b+c is Collinear with d and b+c+d is collinear with a, then a+b+c+d is
(a)0 (b)collinear with a+d
(c)collinear with a-d (d) collinear with b-c

69. Forces of magnitude (5,3,1) units act in the directions $6\hat{\imath} + 2\hat{\jmath} + 3\hat{k}, 3\hat{\imath} - 2\hat{\jmath} + 6\hat{k}, 2\hat{\imath} - 3\hat{\jmath} - 6\hat{k}$ respectively on a particle which is displaced from the point (2,-1,-3) to (5,-1,1). The total work done by the force is
(a)21 units (b)5 units
(c)33 units (d)105 units

70. The position vectors of points A and B are a and b. then the position vectors of point p dividing AB in the ratio m:n is
(a)$\frac{na+mb}{m+n}$ (b) $\frac{na+mb}{m-n}$ (c) $\frac{na-mb}{m+n}$ (d)None

71. If a, b, c are three non-zero vectors with no two of which are collinear, a+2b is collinear with c and b+3c is collinear with a, then$|a + 2b + +6c|$will be equal to
(a)zero (b)9 (c)1 (d)None

72. Vertices of the vectors $\hat{\imath} - 2\hat{\jmath} + 2\hat{k}, 2\hat{\imath} + \hat{\jmath} - \hat{k}$ and $3\hat{\imath} - \hat{\jmath} + 2\hat{k}$ form a triangle, this triangle is
(a)Equilateral triangle (b)Right angle triangle
(c)Two side are equal in length
(d)None of the above

73. If the volume of a parallelopiped whose adjacent edges $a = 2\hat{\imath} + 3\hat{\jmath} + 4\hat{k}, b = \hat{\imath} + a\hat{\jmath} + 2\hat{k}$,$c = \hat{\imath} + 2\hat{\jmath} + a\hat{k}$ is 15, then α is
(a)1 (b)5/2 (c)9/2 (d)0

NIMCET 2021

74. Let $a = 2\hat{\imath} + \hat{\jmath} + 2\hat{k}, b = \hat{\imath} - \hat{\jmath} + 2\hat{k}$ and $c = \hat{\imath} + \hat{\jmath} - 2\hat{k}$ are three vectors. Then, a vector in the plane of a and c whose projection on b is of magnitude $\frac{1}{\sqrt{6}}$ is
(a)$3\hat{\imath} - 2\hat{\jmath}$ (b)$3\hat{\imath} + 2\hat{\jmath}$
(c)$2\hat{\imath} + 3\hat{\jmath} - \hat{k}$ (d)$3\hat{\imath} + 2\hat{\jmath} + \hat{k}$

75. If the position vector of A and B relative to O be $\hat{\imath} - 4\hat{\jmath} + 3\hat{k}$ and $-\hat{\imath} + 2\hat{\jmath} - k$ respectively, then the median through O of ΔOAB is
(a)-$2\hat{\imath} + 2\hat{k}$ (b)$-\hat{\jmath} + \hat{k}$
(c)- $\hat{\imath}$ -$\hat{\jmath}$ +$\hat{k}$ (d)$-\hat{\imath} - \hat{\jmath} - \hat{k}$

76. The area of the triangle formed by the vertices whose position vectors are $3\hat{\imath} + \hat{\jmath}, 5\hat{\imath} + 2\hat{\jmath} + \hat{k}, \hat{\imath} - 2\hat{\jmath} + 3\hat{k}$ is
(a)$\sqrt{21} sq. units$ (b)$\sqrt{23} sq. units$
(c)$\sqrt{33} sq. units$ (d)$\sqrt{29}$ sq. units

77. If the vector $a\hat{\imath} + \hat{\jmath} + \hat{k}, \hat{\imath} + b\hat{\jmath} + \hat{k}, \hat{\imath} + \hat{\jmath} + c$ $(a, b, c \neq 1)$are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} =$
(a)0 (b)1 (c)2 (d)3

78. Let $a = \hat{\imath} + \hat{\jmath}$ and $b = 2\hat{\imath} - \hat{k}$. Then, the point of intersection of the lines $r \times a = b \times a$ and $r \times b = a \times b$ is
(a)$-\hat{\imath} + \hat{\jmath} + \hat{k}$ (b)$3\hat{\imath} - \hat{\jmath} + \hat{k}$
(c)$\hat{\imath} - \hat{\jmath} - \hat{k}$ (d) $3\hat{\imath} + \hat{\jmath} - \hat{k}$

79. Angle between a and b is 120°. If $|b| = 2|a|$ and the vectors a+ xb, a-b are at right angle, then x is equal to
(a)$\frac{1}{3}$ (b)$\frac{1}{5}$ (c)$\frac{2}{3}$ (d)$\frac{2}{5}$

80. If $e_1 = (1, 1, 1)$ and $e_2 = (1,1,-1)$ and a and b are two vectors such that $e_1 = 2a + b$ and $e_2 = a + 2b$, then angle between a and b is
(a)$\cos^{-1}\left(-\frac{7}{11}\right)$ (b)$\cos^{-1}\left(\frac{7}{11}\right)$
(c)$\cos^{-1}\left(\frac{7}{9}\right)$ (d)$\cos^{-1}\left(\frac{6\sqrt{2}}{11}\right)$

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81. If a =$\lambda\hat{\imath}$ +$\hat{\jmath}$ - $2\hat{k}$, b=$\hat{\imath}$ + $\lambda\hat{\jmath}$ -$2\hat{k}$, c = $\hat{\imath}$ + $\hat{\jmath}$ + $\hat{k}$ and [a b c] = 7 , then the values of λ are
(a)2, -6 (b) 4, -2 (c) 4,2 (d)-4,2

82. Let a = $2\hat{\imath} + 2\hat{\jmath} + \hat{k}$ and b be another vector such that a. b = 14 and a× b = $3\hat{\imath} + \hat{\jmath} - 8\hat{k}$ the vector b =
(a)$5\hat{\imath} + \hat{\jmath} + 2\hat{k}$ (b)$5\hat{\imath} - \hat{\jmath} - 2\hat{k}$
(c)$5\hat{\imath} + \hat{\jmath} - 2\hat{k}$ (d)$3\hat{\imath} + \hat{\jmath} + 4\hat{k}$

83. If$(\hat{a} \times \hat{b}) \times \hat{c} = \hat{a} \times (\hat{b} \times \hat{c})$, then
(a)a and b are collinear
(b)a and b are perpendicular
(c)a and c are collinear
(d)a and c are perpendicular

84. Let a, b and c be distinct non-negative number. If the vector $a\hat{\imath} + a\hat{\jmath} + c\hat{k}$, $\hat{\imath} + \hat{k}$ and $c\hat{\imath} + c\hat{\jmath} + b\hat{k}$ lie in a plane then c is equal to
(a) the arithmetic means of a and b
(b) the geometric mean of a and b
(c) the harmonic mean of a and b
(d) equal to zero

85. If the volume of a parallelopiped whose adjacent edges are a = $2\hat{\imath} + 3\hat{\jmath} + 4\hat{k}$, b = $\hat{\imath} + \alpha\hat{\jmath} + 2\hat{k}$, c = $\hat{\imath} + 2\hat{\jmath} + \alpha\hat{k}$is 15,then α is equal to
(a)1 (b)$\frac{5}{2}$ (c)$\frac{9}{2}$ (d)0

NIMCET 2023

86. If a and b are vector in space , given by a = $\frac{\hat{\imath}-2\hat{\jmath}}{\sqrt{5}}$ and $b = \frac{2\hat{\imath}+\hat{\jmath}+3\hat{k}}{\sqrt{14}}$, then the value of $(2a + b).[(a \times b) \times (a - 2b)]$is
(a)3 (b)4 (c)5 (d)6

87. Let $A = 2\hat{\imath} + \hat{\jmath} - 2\hat{k}$ and $B = \hat{\imath} + \hat{\jmath}$. If C is a vector such that $|C - A| = 3$ and the angle between A× B and C is 30° , then $[(A\times B) \times C] = 3$. Then, the value of A . C is equal to
(a)25/8 (b)2 (c)5 (d)1/8

88. *If* $a = \hat{\imath} - \hat{k}, b = x\hat{\imath} + \hat{\jmath} + (1 - x)\hat{k}$ and $c = y\hat{\imath} + x\hat{\jmath} + (1 + x - y)\hat{k}$, then [a b c] depends on
(a)Neither x nor y (b)Only x

(c)Only y (d)Both x and y

89. If a , b are unit vectors such that 2a + b =3 , then which of the following statement is true?
(a)a is parallel to b (b)a is perpendicular to b
(c)a is perpendicular to 2a + b
(d)b is parallel to 2a + b

90. $\emptyset = cos^{-1}\left(\frac{3}{\sqrt{20}}\right)$ is the angle between a=$\hat{\imath}$ -2x$\hat{\jmath}$ +3y$\hat{k}$ and b=x$\hat{\imath}$ +$\hat{\jmath}$+ $y\hat{k}$
Then possible values at (x, y) that lie on locus
(a) (0,1) (b)(1,0) (c)(1,1) (d)(0,0)

91. If a vector having magnitude of 5 units, makes equal angle with each of the three mutually perpendicular axes, then the sum of the magnitude of the projections on each of the axis is
(a)15/3 unit (b)5$\sqrt{3}$ unit
(c) $\frac{15\sqrt{3}}{2}$ units (d)None of these

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92. Let α, β, γ be distinct real numbers. The points with position vectors $\alpha\hat{I} + \beta\hat{\jmath} + \gamma\hat{k}$, $\beta\hat{\imath} + \gamma\hat{\jmath} + \alpha\hat{k}$, $\gamma\hat{\imath} + \alpha\hat{\jmath} + \beta\hat{k}$
(a)are collinear
(b)form an equilateral triangle
(c)form a scalene triangle
(d)form a right angled triangle

93. If $\vec{a}, \vec{b}, \vec{c}$ are three non-collinear vectors such that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$, then find $\vec{a} + \vec{b} + \vec{c}$
(a)0 (b)3$\vec{a}$ (c)3$\vec{b}$ (d)3$\vec{c}$

94. A particle is acted upon by constant forces $4\hat{\imath} + \hat{\jmath} - 3\hat{k}$ and $3\hat{\imath} + \hat{\jmath} - \hat{k}$ which displaced from a point $\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$ to the point $5\hat{\imath} + 4\hat{\jmath} + \hat{k}$. The total work done by the forces in SI units is:
(a)20 (b)50 (c)60 (d)40

95. The direction of velocity of a boat relative to water is $3i + 4j$ and that of water relative to earth is $i - 3j$ what is the magnitude and direction of boat w.r.t earth?
(a)$\sqrt{15}, \tan^{-1}(-1/4)$ North
(b)$\sqrt{17}, \tan^{-1}(1/4)$
(c)$\sqrt{15}, \tan^{-1}(1/2)$
(d)None of these

96. Let α, β, γ be distinct real numbers. The points with position vectors $\alpha\hat{\imath} + \beta\hat{\jmath} + \gamma\hat{k}$, $\beta\hat{\imath} + \gamma\hat{\jmath} + \alpha\hat{k}$, $\gamma\hat{\imath} + \alpha\hat{\jmath} + \beta\hat{k}$

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(a)are collinear
(b)form an equilateral triangle
(c)form a scalene triangle
(d)form a right angled triangle

97. If $\vec{a}, \vec{b}, \vec{c}$ are three non-collinear vectors such that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$, then find $\vec{a} + \vec{b} + \vec{c}$

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(a)0 (b)3$\vec{a}$ (c)3$\vec{b}$ (d)3$\vec{c}$

98. A particle is acted upon by constant factor $4\hat{\imath} + \hat{\jmath} - 3\hat{k}$ and $3\hat{\imath} + \hat{\jmath} - \hat{k}$ which displace it from a point $\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$ to the point $5\hat{\imath} + 4\hat{\jmath} + \hat{k}$. The work done in standard units by the forces is given by:

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(a)25 (b)30 (c)40 (d)15 w

99. The direction of velocity of a boat relative to water is $3i + 4j$ and that of water relative to earth is $i - 3j$ what is the magnitude and direction of boat w.r.t earth?

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(a)$\sqrt{15}, \tan^{-1}(-1/4)$ North
(b)$\sqrt{17}, \tan^{-1}(1/4)$.
(c)$\sqrt{15}, \tan^{-1}(1/2)$
(d)None of these

100.The vector $\vec{A}$= (2x+1) $\hat{\imath}$ + (x^2 – 6y) $\hat{\jmath}$ + (xy^2 +3z) $\hat{k}$ is a
(a) sink field (b) solenoidal field
(c) source field (d) None of these

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101.A man starts at the origin O and walks a distance of 3 units in the north-east direction and then walks a distance of 4 units in the north-west direction to reach the point P, then $\overline{OP}$ is equal to (Vector)
(a) $\frac{1}{\sqrt{2}}(-\hat{\imath} + \hat{\jmath})$ (b) $\frac{1}{2}(\hat{\imath} + \hat{\jmath})$
(c) $\frac{1}{\sqrt{2}}(\hat{\imath} - 7\hat{\jmath})$ (d) $\frac{1}{\sqrt{2}}(-\hat{\imath} + 7\hat{\jmath})$

NIMCET 2024

102.If |F| = 40 N (Newtons), |D| = 3m, and θ =60^0, then the work done by F acting from P to Q is (Vector)
(a) 60$\sqrt{3}$ J (b) 120J
(c) 60$\sqrt{2}$ J (d) 60J

NIMCET 2024

103.The value of m for which volume of the paralleleopiped is 4 cubic units whose three edges are represented by a = mi + j + k, b = i – j + k, c = i + 2j – k is (Vector)
(a) 0 (b) –2
(c) –1 (d) 1

NIMCET 2024

104.How much work does it take to slide a crate for a distance of 25 m along a loading dock by pulling on it with a 180 N force where the dock is at an angle of 45^0 from the horizontal? (Vector)
(a) 3.18198 x 10^3 J
(b) 3.18198 x 10^2 J
(c) 3.4341 x 10^{3}J
(d) 3.4341 x 10^4 J

NIMCET 2024

105.The number of distinct real values of $\lambda^2\hat{\imath} + \hat{\jmath} + \hat{k}$, $\hat{\imath} + \lambda^2\hat{\jmath} + \hat{k}$ and $\hat{\imath} + \hat{\jmath} + \lambda^2\hat{k}$ are coplanar is (Vector)
(a) 1 (b) 2
(c) 3 (d) 6

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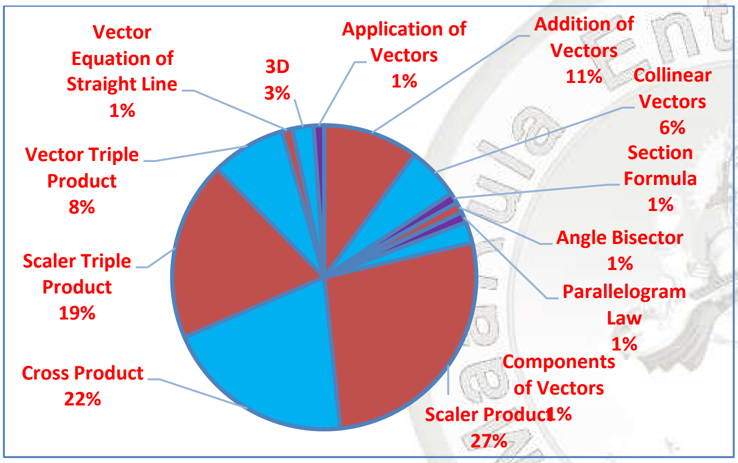
ANSWER

1.	c	2.	b	3.	b	4.	a	5.	c
6.	a	7.	a	8.	b	9.	*	10.	d
11.	d	12.	a	13.	c	14.	c	15.	c
16.	c	17.	d	18.	a	19.	c	20.	b
21.	b	22.	a	23.	c	24.	c	25.	c
26.	b	27.	d	28.	b	29.	b	30.	c
31.	a	32.	a	33.	a	34.	c	35.	d
36.	*	37.	c	38.	a	39.	*	40.	a
41.	a	42.	a	43.	a	44.	c	45.	*
46.	b	47.	d	48.	c	49.	d	50.	a
51.	d	52.	b	53.	d	54.	c	55.	b
56.	d	57.	b	58.	c	59.	d	60.	b
61.	d	62.	d	63.	c	64.	*	65.	b
66.	d	67.	c	68.	a	69.	c	70.	a
71.	a	72.	b	73.	c	74.	b	75.	b
76.	d	77.	b	78.	d	79.	d	80.	a
81.	a	82.	a	83.	c	84.	b	85.	c
86.	c	87.	b	88.	a	89.	a	90.	a
91.	b	92.	b	93.	a	94.	d	95.	b
96.	b	97.	a	98.	c	99.	b	100.	a
101.	d	102.	d	103.	d	104.	d	105.	b

Note(*)wrong.

Analysis 2008 to 2023
Questions Asked in NIMCET 2008 to 2023

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	08	NIMCET 2016	06
NIMCET 2008	05	NIMCET 2017	06
NIMCET 2009	03	NIMCET 2018	04
NIMCET 2010	02	NIMCET 2019	06
NIMCET 2011	05	NIMCET 2020	08
NIMCET 2012	06	NIMCET 2021	07
NIMCET 2013	06	NIMCET 2022	05
NIMCET 2014	06	NIMCET 2023	06
NIMCET 2015	08		



Sub Topic	Qs. Asked
Addition of Vectors	09
Collinear Vectors	05
Section Formula	01
Angle Bisector	01
Parallelogram Law	01
Components of Vectors	02
Scaler Product	24
Cross Product	18
Scaler Triple Product	17
Vector Triple Product	07
Vector Equation of Straight Line	01
3D	02
Application of Vectors	01

SOLUTION

1. **(c)** Volume of parallelepiped = [a, b, c]

$$\begin{vmatrix} 1 & \lambda & 1 \\ 0 & 1 & \lambda \\ \lambda & 0 & 1 \end{vmatrix} = \lambda^3 - \lambda - 1$$

$$f(\lambda) = \lambda^3 - \lambda + 1$$

For minimum $f'(\lambda) = 3\lambda^2 - 1 = 0$

$$\lambda^2 = \frac{1}{3}$$

$$\left(\lambda - \frac{1}{\sqrt{3}}\right)\left(\lambda + \frac{1}{\sqrt{3}}\right)$$

So, clearly $\lambda = \frac{1}{\sqrt{3}}$

2. **(b)** $A = 3\hat{i} + 2\hat{j} + \lambda\hat{k}$, $B = 6\hat{i} + 3\hat{j} + \hat{k}$

$C = 5\hat{i} + 7\hat{j} + 3\hat{k}$ and $D = 2\hat{i} + 2\hat{j} + 6\hat{k}$ are coplanar

$AB = 3\hat{i} + 5\hat{j} + (1-\lambda)\hat{k}$

$BC = -\hat{i} + 4\hat{j} + 2\hat{k}$

$CD = -3\hat{i} - 5\hat{j} + 3\hat{k}$

Now, for coplanar, $\begin{vmatrix} 3 & 5 & (1-\lambda) \\ -1 & 4 & 2 \\ -3 & -5 & 3 \end{vmatrix}$

$\Rightarrow \lambda = 4$

3. **(b)** $A \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = \hat{i}(2) - \hat{j}(2) + \hat{k}(1)$

$A \times B = 2\hat{i} - 2\hat{j} + \hat{k}$

$|C - A| = 2\sqrt{2} \Rightarrow |C - A|^2 = 8$

$|C|^2 + |A|^2 - 2C.A = 8 \Rightarrow |C|^2 + 9 - 2|C| = 8$

$|C|^2 - 2|C| + 1 = 0 \Rightarrow (|C| - 1)^2 = 0$

$\Rightarrow |C| = 1$

Now,

$|(A \times B) \times C| = |A \times B||C| \sin\theta$

$= \sqrt{4+4+1} \times 1 \times \sin 30°$ [$\because$ Given, θ = 30°]

$= \frac{3}{2}$

4. **(a)** $AB = 2\hat{i} - 2\hat{j} + \hat{k}$

$AP = 4\hat{i} + \hat{j} - 2\hat{k}$

$\frac{AB}{|AB|} = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3}$

V of any point P (5, –1, –1) on rigid body rotating about axis AB is given by $V = \left(\left(\frac{AB}{|AB|}\right) \times AP\right).w$

$$\frac{AB}{|AB|} \times AP = \frac{1}{3}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & 1 \\ 4 & 1 & -2 \end{vmatrix} = \frac{3\hat{i} + 8\hat{j} + 10\hat{k}}{3}$$

As, $V = \left(\left(\frac{AB}{|AB|}\right) \times AP\right).w$

Hence, $V = \left(\frac{3\hat{i} + 8\hat{j} + 10\hat{k}}{3}\right)3 = 3\hat{i} + 8\hat{j} + 10\hat{k}$

5. **(c)** A + B = –C

$\Rightarrow |A + B| = C$

On squaring both sides, we get

$|A|^2 + |B|^2 + 2A.B = |C|^2$

$9 + 25 + 2|A||B|\cos\theta = 49$

$2|A||B|(\cos\theta) = 49 - 34$

$\cos\theta = \frac{15}{2}\times\frac{1}{3}\times\frac{1}{5}$

$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{5\pi}{3}$

6. (*a*) According to question,

$B = \lambda A + B_2$

$\left[\begin{array}{l}\because B_1 \text{ is parallel to } A \\ \therefore B_1 = \lambda A\end{array}\right]$

$B_1 = \lambda A$

$3\hat{i} + 4\hat{k} = \lambda(\hat{i}+\hat{j}) + B_2$

Take product with A on both sides

$(3\hat{i}+4\hat{k})(\hat{i}+\hat{j}) = \lambda(\hat{i}+\hat{j})(\hat{i}+\hat{j}) + B_2.A$

$3 = \lambda(2) = 0$

$[\because B_2.A = 0]$

$\Rightarrow \lambda = \frac{3}{2}$

So, $B_1 = \lambda A \Rightarrow B_1 = \frac{3}{2}(\hat{i}+\hat{j})$

7. (*a*) $|a-b|^2 + |b-c|^2 + |c-a|^2$

$= 2(|a|^2+|b|^2+|c|^2) - 2(a.b - 2b.c - 2c.a)$

$= 6 - 2(a.b + 2b.c + 2c.a)$ ...(i)

Now we know that $|a+b+c|^2 > 0$

$|a|^2+|b|^2+|c|^2 + 2(a.b+b.c+c.a) \geq 0$

$3 + 2(a.b+b.c+c.a) \geq 0$

$a.b+b.c+c.a \geq -\frac{3}{2}$

Now, put minimum value of $a.b+b.c+c.a = \frac{3}{2}$ in Eq. (i)

$|a-b|^2 + |b-c|^2 + |c-a|^2$

$= 6 - 2(a.b+b.c+c.a) \leq 6 - 2\left(-\frac{3}{2}\right)$

That is equal to 9.

8. (*b*) $a\times(b\times c) = \frac{b+c}{\sqrt{2}} \Rightarrow |a| = |b| = 1$

$(a.c)\hat{b} - (a.b)\hat{c} = \frac{b}{\sqrt{2}} + \frac{c}{\sqrt{2}}$

Comparing we can say

$-(a.b) = \frac{1}{\sqrt{2}}$

$\Rightarrow ab\cos\theta = \frac{-1}{\sqrt{2}} \Rightarrow \theta = \pi - \frac{\pi}{4} \Rightarrow \theta = \frac{3\pi}{4}$

9. (Wrong) We know from triangle law

PA + AC = PC

PB + BC = PC

Adding PA + AC + PB + BC = 2PC

Now, AC and BC are equal and opposite they will cancel each other.

PA + PB = 2PC

10. (*d*) $a + b = \lambda c$ ... (i)

$b + c = \mu a$ ...(ii)

Eq. (i) – Eq. (ii)

$a - c = \lambda c - \mu a$

$(1+\mu)a = (\lambda+1)c$

But *a* is not parallel to *c*,

Hence $\lambda = -1$ or $\mu = -1$

Put these values in Eq. (i)

$\Rightarrow a + b + c = 0$

11. (*d*) $|a\times b| = |a.b|$

$|a||b|\sin\theta = |a||b|\cos\theta$

$\Rightarrow \sin\theta = \cos\theta$

$\Rightarrow \tan\theta = 1$

$\Rightarrow \theta = \frac{\pi}{4}$

12. (*a*) For three coplanar vectors $[a\ b\ c] = 0$

$\therefore (2a-b).[(2b-c)\times 2c - a]$

$= (2a-b).[(2b\times 2c) - (2b\times a) - 0 + (c\times a)]$

$= \{8[abc] - 0 + 0 - 0 + 0 - [bca]\}$

$= 7[a\ b\ c] = 0$

$[\because [a\ bc] = 0(given)]$

13. (*c*) An angle between *a* and *b* is acute $\cos\theta$ is positive, so *a* . *b* > 0

$(x\hat{i} - 3\hat{j} - \hat{k}).(2x\hat{i} + x\hat{j}) - \hat{k} > 0$

$2x^2 - 3x + 1 > 0$

$2x(x-1) - (x-1) > 0$

$(2x-1)(x-1) > 0$

$x\left(-\infty, \frac{1}{2}\right) \cup (1, \infty)$

Also, $b.j < 0$, so $x < 0$.

14. (*c*) $[uvw] = u.(v\times w)$

$v\times w = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 0 & 3\end{vmatrix} = \hat{i}(3) - \hat{j}(6+1) + \hat{k}(-1) = 3\hat{i} - 7\hat{j} - \hat{k}$

$u.(v\times w) = |u||v\times w|\cos\theta$

Maximum value of $\cos\theta = 1$

As, $|v\times w| = \sqrt{59}$

So, $|\hat{u}||v\times w|\times 1 = \sqrt{59}$

15. (*c*) Using triangle law,

AC = *a* + *b* and BD = *b* – *a*

∴ AC – BD = 2*a* = 2AB

16. (*c*) We know equation of plane in 3D is-

$a(x-x_1) + b(y-y_1) + c(z-z_1) = 0$

Where *a*, *b*, *c* ⇒Direction ratios of vector normal to the plane

As $N = 3\hat{i} - \hat{j} + 2\hat{k}$ is normal to plane

Plane passing through the point (1, 2, 3)

So, $3(x-1) + (-1)(y-2) + 2(x-3) = 0$

$\Rightarrow 3x - y + 2z - 3 + 2 - 6 = 0$

$\Rightarrow 3x - y + 2z - 7 = 0 \Rightarrow 3x - y + 2x = 7$

17. (*d*) $|a+b+c|^2 = |a|^2 + |b|^2 + |c|^2 + 2(a.b+b.c+c.a)$

$0 = 1 + 1 + 1 + 2(a.b+b.c+c.a)$

$a.b+b.c+c.a = -\frac{3}{2}$

18. **(*a*)** For a and b to the perpendicular *a* . *b* = 0
$\left(\hat{i} + x\hat{j} - 2\hat{k}\right).\left(x\hat{i} + 3\hat{j} - 4\hat{k}\right) = 0$
$\Rightarrow \quad x + 3x + 8 = 0 \Rightarrow 4x = -8 \Rightarrow x = -2$

19. **(*a*)** For *a* and *b* to be perpendicular, *a* . *b* = 0
Let $a = \hat{i},\ b = \hat{j},\ c = \hat{k}$
For non-coplanar vectors, $\begin{vmatrix} 1 & 2 & 3 \\ 0 & \lambda & 4 \\ 0 & 0 & 2\lambda - 1 \end{vmatrix} \neq 0$
This must be true
$\lambda(2\lambda - 1) - 0 \neq 0 \Rightarrow \lambda(2\lambda - 1) \neq 0 \Rightarrow 0$ and $\lambda = \frac{1}{2}$
$\lambda \in R$ except true values i.e. 0, 1/2.

20. **(*b*)** $a + b = -c$
$|a + b|^2 = |-c|^2$
$\Rightarrow |a|^2 + |b|^2 + 2a.b = |c|^2$
$\Rightarrow 9 + 25 + 2|a||b|\cos\theta = 49 \Rightarrow |a||b|\cos\theta = \frac{49 - 34}{2}$
$\Rightarrow 15\cos\theta = \frac{15}{2} \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$

21. **(*b*)** $\frac{|a \times b|}{a.b} = \frac{|a||b|\sin\theta}{|a||b|\cos\theta}$
$0 \leq \theta \leq \pi$
Then, $|\sin\theta| = \sin\theta$
Hence, the value of given expression is tan θ .

22. **(*a*)** $(a \times b) + c = 0$
$a \times (a \times b) + a \times c = 0$
$(a\,.\,b)a - (a\,.\,a)b + a \times c = 0$ ...(i)
$a \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & -1 \\ 1 & -1 & -1 \end{vmatrix} = 2\hat{i} - \hat{j} - \hat{k}$
As, $a = \hat{j} - \hat{k}$, $c = \hat{i} - \hat{j} - \hat{k}$
Put these values in Eq. (i)
$3\left(\hat{j} - \hat{k}\right) - 2\hat{i} - \hat{j} - \hat{k} = (a\,.\,a)b$
$2b = -2\hat{i} + 2\hat{j} - 4\hat{k}$
$b = -\hat{i} + \hat{j} - 2\hat{k}$

23. **(*c*)** Volume of parallelepiped = [*a b c*]
Volume = $\begin{vmatrix} 2 & -3 & 0 \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix}$
$= 2(-1) + 3(-1 + 3) + 0 = -2 + 3(2) = -2 + 6 = 4$
Volume = 4

24. **(*c*)** Resultant force = $3\hat{i} + 2\hat{j} + 5\hat{k} + 2\hat{i} + \hat{j} - 3\hat{k}$
$F = 5\hat{i} + 3\hat{j} + 2\hat{k}$
$r = 4\hat{i} - 2\hat{j} + 7\hat{k} - 2\hat{i} + \hat{j} + 3\hat{k} \Rightarrow r = 2\hat{i} - \hat{j} + 10\hat{k}$
W = F . r
$W = \left(5\hat{i} + 3\hat{j} + 2\hat{k}\right)\left(2\hat{i} - \hat{j} + 10\hat{k}\right)$
$W = 10 - 3 + 20 = 27$ units

25. **(*b*)** Area of Parallelogram $= \frac{|BD \times AC|}{2}$
Where BD and AC are diagonals so,
$a \times b = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix} = \hat{i}(4 - 6) - \hat{j}(12 + 2) + \hat{k}(-9 - 1)$
$= -2\hat{j} - 14\hat{j} - 10\hat{k}$
Area $= \frac{|a \times b|}{2} = \frac{-2\hat{i} - 14\hat{j} - 10\hat{k}}{2} = \left|-\hat{i} - 7\hat{j} - 5\hat{k}\right|$
$= \sqrt{1 + 49 + 25} = \sqrt{75} = 5\sqrt{3}$
Area = $5\sqrt{3}$ units

26. **(*d*)** Since, vectors are coplanar
$\therefore \begin{vmatrix} 2 & -1 & 1 \\ 1 & 2 & -3 \\ 3 & \lambda & 5 \end{vmatrix} = 0$
$\Rightarrow 3(3 - 2) - \lambda(-6 - 1) + 5(4 + 1) = 0$ [Expand along R_3]
$\Rightarrow 3 + 7\lambda + 25 = 0$
$\Rightarrow 7\lambda = -28$
$\Rightarrow \lambda = -4$

27. **(*b*)** Area $= \frac{1}{2}|a \times b + b \times c + c \times a|$

28. **(*b*)** $F = 2\hat{i} - 5\hat{j} + 6\hat{k} + \left(-\hat{i}\right) + 2\hat{j} - \hat{k}$
$F = \hat{i} - 3\hat{j} + 5\hat{k}$
$d = \left(6\hat{i} - 4\hat{j}\right) + \left(\hat{j} + 3\hat{j}\right) + \left(-3\hat{k} + 2\hat{k}\right)$
$d + 2\hat{i} + 4\hat{j} - \hat{k}$
$W = F.d$
$W = 2 - 12 - 5 = -10 - 5$
= −15 units

29. **(*c*)** According to question,
$(-4m + 2n)\hat{i} + (2m + n)\hat{j} = 2\hat{i} + 3\hat{j}$
$-4m + 2n = 2$ ... (i)
$2m + n = 3$ ... (ii)
On solving Eqs. (i) and (ii), we get
$m = \frac{1}{2},\quad n = 2$
$m + n = 1/2 + 2 = 5/2$

30. **(*a*)** For right handed system *A* × *B* = *N*
$A \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 3 & 1 \\ 2 & -1 & 2 \end{vmatrix}$
$= \hat{i}(6 + 1) - \hat{j}(8 - 2) + \hat{k}(-4 - 6)$
$= 7\hat{i} - 6\hat{j} - 10\hat{k}$
Since N is unit vector $N = \frac{N}{|N|}$
$|N| = \sqrt{49 + 36 + 100} = \sqrt{185}$
$N = \frac{7\hat{i} - 6\hat{j} - 10\hat{k}}{\sqrt{185}}$

31. **(*a*)** $a + b = c$

$|a+b|^2 = |c|^2$

$|a|^2 + |b|^2 + 2a.b = |c|^2$

$2a.b = -4$

Now, $|a-b|^2 = |a|^2 + |b|^2 - 2a.b$

$|a-b|^2 = 4 + 4 - (-4)$

$|a-b|^2 = 12$

$|a-b| = \sqrt{12}$

$\Rightarrow |a-b| = 2\sqrt{3}$

32. (*a*) Vector is making θ angle with coordinate axes so,

For $l\hat{i} + m\hat{j} + n\hat{k}$

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Since, angles are same,

$\cos^2\theta + \cos^2\theta + \cos^2\theta = 1$

$3\cos 2\theta = 1$

$\Rightarrow \quad \cos^2\theta = 1/3$

$\cos\theta = \dfrac{1}{\sqrt{3}}$

$\Rightarrow \theta = \cos^{-1}\dfrac{1}{\sqrt{3}}$

33. (*b*) Let A, B and C have position vectors a, b and c.

Now, assume $BE : EF = \mu : 1$

$AF : FC = \lambda : 1$

Now, $OE = \dfrac{b+c}{4}$

$OF = \dfrac{\lambda c + a}{\lambda + 1}$

$\dfrac{b+c}{4} = \dfrac{\mu\left(\dfrac{\lambda c + a}{\lambda + 1}\right) + b}{\mu + 1}$

$(b+c)(\mu+1)c = 4\left[\mu\left(\dfrac{\lambda c + a}{\lambda + 1}\right) + b\right]$

$b(\mu+1) + (\mu+1)c = \dfrac{4\mu(\lambda c + a)}{\lambda + 1} + 4b$

Comparing coefficients of *b* and *c*, we get

$\mu + 1 = 4;\ \mu = 3$

$b(\mu+1) + c(\mu+1) = \dfrac{4\mu\lambda c}{\lambda+1} + \dfrac{4\mu\lambda a}{\lambda+1} + 4b$

$\Rightarrow \mu + 1 = \dfrac{4\mu\lambda}{\lambda+1}$

Putting $\mu = 3$

$3 + 1 = \dfrac{4\times 3\lambda}{\lambda+1} \Rightarrow 4 = \dfrac{12\lambda}{\lambda+1} \Rightarrow 1 = \dfrac{3\lambda}{\lambda+1}$

$\Rightarrow \lambda + 1 = 3\lambda$

$\Rightarrow 2\lambda = 1 \Rightarrow \lambda = \dfrac{1}{2}$

$\dfrac{AF}{FC} = \dfrac{1}{2}$

34. (*d*) $|a\times b| = |a||b|\sin\theta$

$a.b = |a||b|\cos\theta$

$\Rightarrow$ 12=13 × cosθ $\Rightarrow$ cosθ= $\dfrac{12}{13}$

$\Rightarrow \quad \sin\theta = \dfrac{5}{13}$

$|a\times b| = 13\times 5\times\dfrac{5}{13} = 25$

35. (***Wrong***) Component of a along b = $\left(\dfrac{a.b}{|b|}\right)\hat{b}$

$\hat{b} = \dfrac{3\hat{i} + 4\hat{k}}{5}$

Component = $\left(\dfrac{12}{5}\right)\dfrac{3\hat{i}+4\hat{k}}{5} = \dfrac{12}{25}(3\hat{i} + 4\hat{k})$

36. (*c*) $-(2a+b).[(a-2b)\times(a\times b)]$

$= -(2a+b) - [\{(a-2b).b\}a - \{(a-2b).a\}b]$

$= (2a+b)[(a-2b).a]b - \{(a-2b)-b\}a]$

$= (2a+b)[\{|a|^2 - 2b.a\}b - \{a.b - 2|b|^2\}a]$

Now, $a.b = \left(\dfrac{\hat{i} - 2\hat{j}}{\sqrt{5}}\right).\left(\dfrac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}\right) = 2 - 2 = 0$

$|a| = |b| = 1$

$2a + b.[\{1-0\}b - \{0-2\}a]$

$= (2a+b).[b+2a] = (2a+b).[b+2a]$

$= 2a.b + 4|a|^2 + |b|^2 + 2a.b = 0 + 4 + 1 + 0 = 5$

37. (*a*) $\begin{vmatrix} 1 & 0 & -1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix}$

$= \{(1+x-y)(x-x^2) - 0 - (x^2 - y)\}$

$= 1 + x - y - x + x^2 - x^2 + y = 1$

Depends neither *x* nor *y*.

38. (***Wrong***) In option (*b*), (*a* + *b*) and (*a* × *b*) are mutually perpendicular.
In option (*d*), (*a* – *b*) and (*a* × *b*) are mutually perpendicular.
In option (*a* and *c*), (*a* + *b*) and *a*, (*a* – *b*) vectors may or may not be perpendicular to each other.
So, nothing can be said in this question.

39. (*a*) Let $v = a + \lambda b$

$v = \hat{i} + \hat{j} + \hat{k} + \lambda\hat{i} - \lambda\hat{j} + \lambda\hat{k}$

$v = (1+\lambda)\hat{i} + (1-\lambda)\hat{j} + (1+\lambda)\hat{k}$

$\dfrac{c}{|c|} = \dfrac{\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}$

$v.\hat{c} = \{(1+\lambda)\hat{i} + (1-\lambda)\hat{j} + (1+\lambda)\hat{k}\}\dfrac{(\hat{i} - \hat{j} - \hat{k})}{\sqrt{3}}$

$\dfrac{1}{\sqrt{3}} = \dfrac{(1+\lambda) - (1-\lambda) - (1+\lambda)}{\sqrt{3}}$

$\lambda = 2$

$v = 3\hat{i} - \hat{j} + 3\hat{k}$

40. (*a*) We have *a*, *b* and *c* are the position vectors of the vertices of A, B and C of a ΔABC.

$= \dfrac{1}{2}(b-a)\times(c\times a)$

$\dfrac{1}{2}|a\times b + b\times c + c\times a|$

41. (**a**) Area of quadrilateral $= \dfrac{|AC\times BD|}{2}$

$$AC \times BD = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ -1 & 3 & 2 \end{vmatrix} = \hat{i}(-1) - \hat{j}(4+1) + \hat{k}(6+1)$$

$$= -\hat{i} - 5\hat{j} + 7\hat{k}$$

Area $= \dfrac{\left|-\hat{i} - 5\hat{j} + 7\hat{k}\right|}{2}$

$$= \frac{5\sqrt{3}}{2}$$

42. **(*a*)** By result we know PA + PB = 2PC

43. **(*c*)** Volume of parallelepiped is [U V W]

Volume $= \begin{vmatrix} 1 & 2 & -1 \\ -2 & 0 & 3 \\ 0 & 7 & -4 \end{vmatrix} = -23$

Volume $= |-23| = 23$ units

44. **(*Wrong*)**

Let $A = \hat{i} - \hat{j}$, $B = -2\hat{i} + \hat{j} - \hat{k}$, $C = -\hat{i} + \hat{j} + 2\hat{k}$

$AB = -3\hat{i} + 2\hat{j} - \hat{k}$, $BC = \hat{i} + 3\hat{k}$

$$AB \times BC = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 2 & -1 \\ 1 & 0 & 3 \end{vmatrix} = 6\hat{i} + 8\hat{j} + (-2)\hat{k}$$

$$AB \times BC = 6\hat{i} + 8\hat{j} - 2\hat{k}$$

45. **(*b*)** $|a+b|^2 = |a|^2 + |b|^2 + 2ab$

$\Rightarrow \alpha^2 = \alpha^2 + \alpha^2 + 2a.b \Rightarrow 2a.b = -\alpha^2$

$|a-b|^2 = |a|^2 + |b|^2 - 2a.b$

$\Rightarrow |a-b|^2 = \alpha^2 + \alpha^2 + \alpha^2 = 3\alpha^2 \Rightarrow |a-b| = \sqrt{3}\alpha$

46. **(*d*)** $a + 2b = \lambda c$ ...(i)

$b + 3c = \mu a$ ...(ii)

Eq. (i) – 2 × Eq. (ii)

$a - 6c = \lambda c - 2\mu a \Rightarrow (1+2\mu)a = (\lambda+6)c$

As we know that a and c is non collinear vectors,
Hence, But no two are collinear, so it means

$\lambda + 6 = 0 \Rightarrow \lambda = -6$

$1 + 2\mu = 0 \Rightarrow \mu = -\dfrac{1}{2}$

Put these values in Eq. (i)

$\Rightarrow a + 2b = -6c \Rightarrow a + 2b + 6c = 0 \Rightarrow |a + 2b + 6c| = 0$

47. **(*c*)** Given, $|a+b| = \sqrt{29}$

And $a \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times b$

$a \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = -b \times (2\hat{i} + 3\hat{j} + 4\hat{k})$

$\Rightarrow (a+b) \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = 0$

$\Rightarrow a + b = \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}) \Rightarrow |a+b| = |\lambda|\sqrt{4+9+16}$

$\Rightarrow \sqrt{29} = |\lambda|\sqrt{29} \Rightarrow |\lambda| = 1 \Rightarrow \lambda = \pm 1$

$\therefore a + b = \pm(2\hat{i} + 3\hat{j} + 4\hat{k})$

Hence, $(a+b).(-7\hat{i} + 2\hat{j} + 3\hat{k})$

$\Rightarrow (2\hat{i} + 3\hat{j} + 4\hat{k})(-7\hat{i} + 2\hat{j} + 3\hat{k}) \Rightarrow -14 + 6 + 12 = 4$

48. **(*d*)** $|a| = \sqrt{4+1+25} = \sqrt{30}$

Direction cosines $= \dfrac{-2}{\sqrt{30}}, \dfrac{1}{\sqrt{30}}, \dfrac{-5}{\sqrt{30}}$

49. **(*a*)** $b + c = -a$

$|b+c|^2 = |a|^2$

$|b|^2 + |c|^2 + 2b.c = |a|^2 \Rightarrow 34 + 2 \times 5 \times 3\cos\theta = 49$

$\Rightarrow 30\cos\theta = 49 - 34 \Rightarrow \cos\theta = \dfrac{15}{30}$

$\Rightarrow \cos\theta = \dfrac{1}{2} \Rightarrow \theta = \cos^{-1}\left(\dfrac{1}{2}\right) = 60° \Rightarrow \theta = 60°$

50. **(*d*)** For coplanar, non zero vectors

$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

$C_2 \to C_2 - C_1$ and $C_3 \to C_3 \to C_1$

$$\begin{vmatrix} a & 1-a & 1-a \\ 1 & b-1 & 1 \\ 1 & 1 & c-1 \end{vmatrix} = 0$$

$\Rightarrow a(b-1)(c-1) - (1-a)(c-1) + (1-a)(1-b) = 0$

$\Rightarrow a(1-b)(1-c) - (1-a)(1-c) + (1-a)(1-b) = 0$

Divided by $(1-a)(1-b)(1-c)$

$$\frac{a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 0$$

Now, the value of

$$\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = \frac{1}{1-a} - \frac{a}{1-a} = 1$$

51. **(*b*)** Given, $|a| = |b| = 1, |c| = 2$

$a \times (a \times c) - b = 0$

$(a.c)a - (a.a)c = b$

$(a.c)a - |a|^2 c = b$

$|(a.c)a - c|^2 = |b|^2$

$\Rightarrow |a.c|^2|a|^2 + |c|^2 - 2(a.c)(a.c) = 1$

$\Rightarrow |a.c|^2 + 4 - 2|a.c|^2 = 1 \Rightarrow 4 - |a.c|^2 = 1$

$\Rightarrow 4 - |a|^2|c|^2\cos^2\theta = 1$

$\Rightarrow 3 = 4\cos^2\theta$

$\Rightarrow \cos^2\theta = \dfrac{3}{4} \Rightarrow \cos\theta = \pm\dfrac{\sqrt{3}}{2} \Rightarrow \theta = \dfrac{\pi}{6}$

52. **(*d*)** We know

$|a+b+c|^2 = |a|^2 + |b|^2 + |c|^2 + 2(a.b + b.c + c.a)$

$0 = 4 + 9 + 25 + 2(a.b + b.c + c.a)$

$\Rightarrow \dfrac{-(4+9+25)}{2} = a.b + b.c + c.a$

$\Rightarrow a.b + b.c + c.a = -\dfrac{38}{2} = -19$

53. **(*c*)** $a + b = 4\hat{i} + \hat{j} - \hat{k}$

$a - b = -2\hat{i} + 3\hat{j} - 5\hat{k}$

$(a+b).(a-b) = |a+b||a-b|\cos\theta$

$-8 + 3 + 5 = \sqrt{16+1+1}\sqrt{4+9+25}\cos\theta$

$\Rightarrow 0 = \cos\theta \Rightarrow \theta = \dfrac{\pi}{2}$

54. **(c)** Refer to the Solution 7.

55. **(d)** $a = b + \lambda c$

$\alpha\hat{i} + 2\hat{j} + \beta\hat{k} = \hat{i} + \hat{j} + \lambda\hat{j} + \lambda\hat{k}$

$\alpha\hat{i} + 2\hat{j} + \beta\hat{k} = \hat{i} + (1+\lambda)\hat{j} + \lambda\hat{k}$

$1 + \lambda = 2$

$\lambda = 1, \alpha = 1$ so, $\beta = 1$

$\alpha = \beta = 1$

56. **(b)** $F_{net} = 4\hat{i} - 3\hat{j} + 7\hat{k} + 2\hat{i} + 2\hat{j} - 8\hat{k} = 6\hat{i} - \hat{j} - \hat{k}$

$r = 2\hat{i} + 5\hat{j} - 6\hat{k} - 5\hat{i} - 7\hat{j} - \hat{k} = -3\hat{i} - 2\hat{j} - 7\hat{k}$

$W = F \cdot r$

$W = -9$

$W = 9$ units.

57. **(c)** Here, velocity vector of bird is

$v = 10\hat{i} + 6\hat{j} + \hat{k}$

Or $|v_x| = 10\,km/h$

$|v_y| = 6\,km/h$

$|v_z| = 1\,km/h$

Initial position vector of the bird

$OA = \hat{i} + 2\hat{j} + 13\hat{k}$

$AB = OB - OA = 10\hat{k}$

$\therefore |AB| = 10\,m$

When bird reached a point in space that is 13m high from ground then final position vector of the bird will be

$OB = \hat{i} + 2\hat{j} + 13\hat{k}$

$AB = OB - OA = 10\hat{k}$

$\therefore \quad |AB| = 10\,m$

Time taken by bird to cover a distance of 10m along the vertical is-

$$t = \frac{AB}{V_z} = \frac{10}{1 \times \frac{5}{18}} = \frac{10 \times 18}{5} = 36 \text{ sec}$$

58. **(b)** $a \cdot b = 1$

$|a||b|\cos\theta = 1 \Rightarrow 2|b|\cos\frac{\pi}{3} = 1$

$\Rightarrow \quad |b|\cos\frac{\pi}{3} = \frac{1}{2} \Rightarrow |b|\frac{1}{2} = \frac{1}{2} \Rightarrow |b| = 1$

Given, $(r + 2a - 10b) = \lambda(a \times b)$

$(r + 2a - 10b)\cdot(a \times b) = 6$ ...(i)

Putting $r + 2a - 10b = \lambda(a \times b)$ in Eq. (i), we get

$\lambda(a \times b)\cdot(a \times b) = 6$

$\lambda|a \times b|^2 = 6$

$\lambda|a|^2|b|^2\sin^2\theta = 6$

$4\lambda\sin^2\frac{\pi}{3} = 6$

$4\lambda\left(\frac{\sqrt{3}}{2}\right)^2 = 6 \Rightarrow \lambda = 2$

59. **(d)** $|ED| = |DE| = 4$ c

60. **(d)** ABCD is a parallelogram and P is mid-point of AD. BP and AC intersect at Q.

Let AQ : QC = λ : 1 and PQ : QB = μ : 1

Assume A as origin B has PV b, C has PV c and D has d. According to parallelogram law c = b + d

PV of $Q = \frac{\lambda c}{\lambda + 1}$

PV of Q for line BP

$$Q = \frac{\mu b + 1 \times d/2}{\mu + 1}$$

$$\frac{\lambda c}{\lambda + 1} = \frac{\lambda(b+d)}{\lambda + 1} = \frac{\mu}{\mu + 1}b + \frac{d}{2(\mu + 1)}$$

Equating these two vectors

$$\frac{\lambda}{\lambda+1} = \frac{\mu}{\mu+1} \text{ and } \frac{\lambda}{\lambda+1} = \frac{1}{2(\mu+1)}$$

Hence, $\frac{\mu}{\mu+1} = \frac{1}{2(\mu+1)}$

Hence, $\mu = \frac{1}{2}$

$\frac{\lambda}{\lambda+1} = \frac{1/2}{3/2} = \frac{1}{3}$

Hence, $\lambda = \frac{1}{2}$

AQ : QC = 1 : 2

61. **(c)** Refer to Solution 33.

62. **(Wrong)** $\alpha(a + 2b) - \beta(4b - a) = 0$

$\Rightarrow \quad (\alpha + \beta)a + (2\alpha - 4\beta)b = 0$

$\Rightarrow \quad \alpha + \beta = 0$ and $2\alpha - 4\beta = 0$

$\Rightarrow \quad \alpha + \beta = 0$ and $\alpha - 2\beta = 0 \qquad \Rightarrow \alpha = \beta = 0$

63. **(b)** Force at point $(r) = 2\hat{i} + 3\hat{j} + 5\hat{k}$

Force is acting along $= \frac{2\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{9}} = \frac{2\hat{i} + 2\hat{j} + \hat{k}}{3}$

So, force will be $= 78\left(\frac{2\hat{i} + 2\hat{j} + \hat{k}}{3}\right) = 26(2\hat{i} + 2\hat{j} + \hat{k})$

We know moment of force (τ) is $\tau = r \times F$

$\tau = (2\hat{i} + 3\hat{j} + 5\hat{k}) \times (26)(2\hat{i} + 2\hat{j} + \hat{k})$

$$\tau = 26\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 5 \\ 2 & 2 & 1 \end{vmatrix} = (26)\left[\hat{i}(-7) + \hat{j}(-8) + \hat{k}(-2)\right]$$

$\tau = 26\left[-7\hat{i} + 8\hat{j} - 2\hat{k}\right]$

Now its magnitude along time joining $12\hat{i} + 3\hat{j} + 4\hat{k}$ with origin will be

Moment's magnitude $= 26\left[-7\hat{i} + 8\hat{j} - 2\hat{k}\right]\frac{(12\hat{i} + 3\hat{j} + 4\hat{k})}{\sqrt{144 + 9 + 16}}$

$= 26\left[-7\hat{i} + 8\hat{j} - 2\hat{k}\right]\frac{(12\hat{i} + 3\hat{j} + 4\hat{k})}{13}$

$= 2(-84 + 24 - 8) = 2(-60 - 8) = 2(-68) = -136$

64. **(d)** $(a + b + c)[(a \times a) + (a \times c) + (b \times a) + (b \times c)]$

$= (a + b + c).[(a \times c) + (b \times a) + (b \times c)]$

$= [a\,bc] - [a\,bc] - [a\,bc] = -[a\,bc]$

65. **(c)** Resultant of two forces $= \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$

So, according to question,

$\sqrt{F_1^2+F_2^2}=\sqrt{F_1^2+F_2^2+2F_1F_2\cos\theta}$

$\Rightarrow F_1^2+F_2^2=F_1^2+F_2^2+2F_1F_2\cos\theta$

$\Rightarrow \cos\theta=0 \Rightarrow \theta=\frac{\pi}{2}$

66. **(a)** $a+b+c=\lambda d,\ b+c+d=\mu a,\ d=\mu a-b-c$

$a+b+c=\lambda(\mu a-b-c)$

$a+b+c=\lambda\mu a-\lambda b-\lambda c$

Comparing coefficients, we get

$\lambda\mu=1,\ \lambda=-1$, then $\mu=-1$

$a+b+c=\lambda d \Rightarrow a+b+c=-d \Rightarrow a+b+c+d=0$

67. **(c)** W = Fd

$d=3\hat{i}+4\hat{k}$

$F=\frac{5(6\hat{i}+2\hat{j}+3\hat{k})}{7}+\frac{3(3\hat{i}+2\hat{j}+6\hat{k})}{7}+\frac{(2\hat{i}+3\hat{j}+6\hat{k})}{7}$

$W=\frac{[41\hat{i}+\hat{j}+27\hat{k}]}{7}.(3\hat{i}+4\hat{k})=\frac{1}{7}(123+108)=\frac{231}{7}=33$

W = 33 units

68. **(a)** Using section formula

$P=\frac{mb+na}{m+n}$

69. **(a)** Refer to solution 46.

70. **(b)** $OA=\hat{i}-2\hat{j}+2\hat{k};\ OB=2\hat{i}+\hat{j}-\hat{k};\ OC=3\hat{i}-\hat{j}+2\hat{k}$

$AB=\hat{i}+\hat{j}-3\hat{k};\ |AB|=\sqrt{19}$

$BC=\hat{i}-2\hat{j}+3\hat{k};\ |BC|=\sqrt{14}$

$\Rightarrow CA+2\hat{i}+\hat{j};\ |CA|=\sqrt{5}$

$|CA|^2+|BC|^2=|AB|^2$

Right angle triangle.

71. **(c)** Volume of parallelepiped $=[a\ b\ c]$

$\begin{bmatrix}2&3&4\\1&\alpha&2\\1&2&\alpha\end{bmatrix}=15$

$\Rightarrow 2(\alpha^2-4)-3(\alpha-2)+4(2-\alpha)=15$

$\Rightarrow 2\alpha^2-7\alpha-9=0$

$\Rightarrow \alpha=\frac{9}{2},-1$

72. **(b)** 2OD = OA + OB

$OD=\frac{-2\hat{j}+2\hat{k}}{2}=-\hat{j}+\hat{k}$

73. **(d)** Point of intersection = *r*

$r\times a=b\times a$ ... (i)

$r\times b=a\times b$ ... (ii)

Adding Eqs. (i) and (ii)

$r\times a+r\times b=b\times a+a\times b$

$\because [b\times a=-a\times b]$

$\Rightarrow r\times(a+b)=0$

$[r\,||\,(a+b)]$

$r=\lambda(a+b)$

Put $r=\lambda(a+b)$ in Eqs. (i)

$(\lambda a+\lambda b)\times a=b\times a$

$[\lambda(b\times a)]=[b\times a]$

$[\because a\times a=0]$

$\lambda=1$

$r=(a+b)$

$r=(\hat{i}+\hat{j})+(2\hat{i}-\hat{k}) \Rightarrow r=3\hat{i}+\hat{j}-\hat{k}$

74. **(b)** Refer to solution 50.

75. **(d)** $(a+xb).(a-b)=0$

$|a|^2-a.b+xa.b-x|b|^2=0$

$|a|^2+a.b(x-1)-4x|a|^2=0$

$|a|^2(1-4x)+a.b(x-1)=0$

$a.b(x-1)=(4x-1)|a|^2$

$\Rightarrow |a||b|\cos 120°(x-1)=(4x-1)|a|^2$

$\Rightarrow 2\left(\frac{-1}{2}\right)(x-1)=(4x-1)\Rightarrow 1-x=4x-1\Rightarrow 2=5x$

$\Rightarrow x=2/5$

76. **(c)** $AB=2\hat{i}+\hat{j}+\hat{k}$ and $AC=-2\hat{i}-3\hat{j}+3\hat{k}$

Area of triangle $=\frac{1}{2}|AB\times AC|$

$AB\times AC=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&1\\-2&-3&3\end{vmatrix}$

$=\hat{i}(3+3)-\hat{j}(6+2)+\hat{k}(-6+2)=6\hat{i}-8\hat{j}-4\hat{k}$

Area $=|-3\hat{i}-4\hat{j}-2\hat{k}|=\sqrt{9+16+4}=\sqrt{29}$ units.

77. **(a)** $e_1=2a+b\ e_2=a+2b$

$\Rightarrow 2e_1=4a+2b\Rightarrow e_2=a+2b$

$2e_1-e_2=3a$

$\Rightarrow a=\frac{2e_1-e_2}{3}$

$e_2=\frac{2e_1-e_2}{3}+2b$

$\Rightarrow 2b=e_2-\frac{2e_1-e_2}{3}$

$=\frac{3e_2-(2e_1-e_2)}{3}=\frac{4e_2-2e_1}{3}$

$b=\frac{2e_2-e_1}{3}$

Now, $a=\frac{2(\hat{i}+\hat{j}+\hat{k})-(\hat{i}+\hat{j}-\hat{k})}{3}=\frac{\hat{i}+\hat{j}+3\hat{k}}{3}$

$b=\frac{2(\hat{i}+\hat{j}-\hat{k})-(\hat{i}+\hat{j}-\hat{k})}{2}=\frac{\hat{i}+\hat{j}-3\hat{k}}{3}$

$a.b=|a||b|\cos\theta$

$\Rightarrow \frac{(1+1-9)}{9}=\frac{\sqrt{11}}{3}\frac{\sqrt{11}}{3}\cos\theta$

$\Rightarrow -7=11\cos\theta$

$\Rightarrow \cos\theta=\frac{-7}{11}\Rightarrow\theta=\cos^{-1}\left(\frac{-7}{11}\right)$

78. **(b)** Let vector to $V=(2\hat{i}+\hat{j}+2\hat{k})+\lambda(\hat{i}+\hat{j}-2\hat{k})$

$V=(2+\lambda)\hat{i}+(1+\lambda)\hat{j}+(2-2\lambda)\hat{k}$

Using projection formula,

$\frac{V.b}{|b|}=\frac{1}{\sqrt{6}}$

$$\Rightarrow \frac{2+\lambda-1-\lambda+4-4\lambda}{\sqrt{6}} = \frac{1}{\sqrt{6}}$$

$\Rightarrow 5-4\lambda = 1 \Rightarrow 4\lambda = 4$

$\Rightarrow \lambda = 1$

$\Rightarrow v = 3\hat{i} + 2\hat{j}$

79. **(a)** $\Rightarrow \begin{vmatrix} \lambda & 1 & -2 \\ 1 & \lambda & -2 \\ 1 & 1 & 1 \end{vmatrix} = 7$

Expand along R_3,

$1(-2+2\lambda) - 1(-2\lambda+2) + 1(\lambda^2-1) = 7$

$\Rightarrow 5 + 4\lambda + \lambda^2 = 7 \Rightarrow \lambda^2 + 4\lambda - 12 = 0$

$\Rightarrow (\lambda+6)(\lambda-2) = 0 \Rightarrow \lambda = -6, 2$

80. **(*a*)** Given, $a = 2\hat{i} + 2\hat{j} + \hat{k},\ a.b = 14,\ a \times b = 3\hat{i} + \hat{j} - 8\hat{k}$

$a \times b = 3\hat{i} + \hat{j} - 8\hat{k}$

$a \times (a \times b) = a \times (3\hat{i} + \hat{j} - 8\hat{k})$

$$a \times (a \times b) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 3 & 1 & -8 \end{vmatrix} = -17\hat{i} + 19\hat{j} - 4\hat{k}$$

$\Rightarrow (a.b)a - (a.a)b = -17\hat{i} + 19\hat{j} - 4\hat{k}$

$\Rightarrow 14(2\hat{i} + 2\hat{j} + \hat{k}) + 17\hat{i} - 19\hat{j} + 4\hat{k} = (a.a)b$

$\Rightarrow 9b = 45\hat{i} + 9\hat{j} + 18\hat{k} \Rightarrow b = 5\hat{i} + \hat{j} + 2\hat{k}$

81. **(*c*)** $(\hat{a} \times \hat{b})\hat{c} = \hat{a} \times (\hat{b} \times \hat{c})$

$(\hat{c}.\hat{a})\hat{b} - (\hat{c}.\hat{b})\hat{a} = (\hat{a}.\hat{c})\hat{b} - (\hat{a}.\hat{b})\hat{c}$

$(\hat{c}.\hat{b})\hat{a} = (\hat{a}.\hat{b})\hat{c}$

$\lambda\hat{a} = \mu\hat{c}$

$a \parallel c$

$$\left\{ \begin{array}{c} -\hat{c} \times (\hat{a} \times \hat{b}) \\ -[(\hat{c}.\hat{b})\hat{a} - (\hat{c}.\hat{a})\hat{b}] \\ (\hat{c}.\hat{a})\hat{b} - (\hat{c}.\hat{b})\hat{a} \end{array} \right\}$$

$\hat{a}$ and $\hat{c}$ are collinear.

82. **(*b*)** All vectors line in a plane.

$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$

Expand along R_2,

$\Rightarrow -1(ab-c^2) + 0(ab-c^2) - 1(ac-ac) = 0$

$\Rightarrow -ab + c^2 = 0 \Rightarrow c^2 = ab$

$a, c, b \to G.P.$

$\therefore$ *c* is geometric mean of *a* and *b*.

83. **(*c*)** $\begin{vmatrix} 2 & 3 & 4 \\ 1 & \alpha & 2 \\ 1 & 2 & \alpha \end{vmatrix} = 15$

$\Rightarrow 2(\alpha^2-4) - 3(\alpha-2) + 4(2-\alpha) = 15$

$\Rightarrow 2\alpha^2 - 8 - 3\alpha + 6 + 8 - 4\alpha = 15$

$\Rightarrow 2\alpha^2 - 7\alpha - 9 = 0 \Rightarrow 2\alpha^2 - 9\alpha + 2\alpha - 9 = 0$

$\Rightarrow \alpha(3\alpha - 9) + 1(2\alpha - 9) = 0$

$\Rightarrow \alpha = \frac{9}{2}, -1$

84. **(*c*)** Refer to solution 36.

85. **(*b*)** $A \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = \hat{i}(2) - \hat{j}(2) + \hat{k}(1)$

$A \times B = 2\hat{i} - 2\hat{j} + \hat{k}$

$|C - A| = 3$

$|C - A|^2 = 9$

$|C|^2 + |A|^2 - 2C.A = 9$

$\Rightarrow |C|^2 - 2C.A = 0$ ...(i)

Now, $|(A \times B) \times C| = |A \times B||C| \sin\theta$

$= \sqrt{4+4+1} \times |C| \times \sin 30° = 3$

Given $\theta = 30°$

$\Rightarrow |C| = 2$

Put this value in Eq. (i)

We get C.A = 2

86. **(*a*)** Refer to solution 37.

87. **(*a*)** Fundamentally this question is wrong because sum of two vectors can not be equal to scalar. But if we have to solve this equation,

$|2a + b|^2 = 9$

$\Rightarrow 4|a|^2 + |b|^2 + 4a.b = 9 \Rightarrow a.b = 1$

$\Rightarrow |a||b|\cos\theta = 1$

$\Rightarrow \cos\theta = 1$

$\Rightarrow \theta = 0$

88. **(*a*)** $a = \hat{i} - 2x\hat{j} + 3y\hat{k}$ and $b = x\hat{i} + \hat{j} + y\hat{k}$

$$\cos\theta = \frac{a.b}{|a||b|} \Rightarrow \cos\cos^{-1}\left(\frac{3}{\sqrt{20}}\right) = \frac{a.b}{|a||b|}$$

$$\Rightarrow \frac{3}{\sqrt{20}} = \frac{x - 2x + 3y^2}{\sqrt{1+4x^2+9y^2}\ \sqrt{x^2+1+y^2}}$$

$\Rightarrow 3\sqrt{1+4x^2+9y^2}\ \sqrt{x^2+1+y^2} = \sqrt{20}\,[3y^2 - x]$

From given option (0, 1) satisfy given equations.

89. **(*b*)** $|v| = 5$

As *v* makes equal angle with all the three axis,

$v = x\hat{i} + x\hat{j} + x\hat{k}$

$|v| = \sqrt{3}x = 5$

$\Rightarrow x = \frac{5}{\sqrt{3}}$

Sum of component on *x*, *y*, *z* axis

$$x + x + x = 3\left(\frac{5}{\sqrt{3}}\right) = 5\sqrt{3}$$

100.(a)

A= $(2x+1)\hat{\imath} + (x^2-6y)\hat{\jmath} + (xy^2+3z)$ k

∇. A = 0 $\to$ Solenoidal

∇. A =-ve $\to$ sink

∇. A =+ve $\to$ source

∇.A $= \left(\frac{\partial}{\partial x} i\right) + \left(\frac{\partial}{\partial y} j +\right) + \left(\frac{\partial}{\partial z} k\right). (2x+1)\hat{\imath} + (x^2-6y)\hat{\jmath} + (xy^2+3z)k$

$\frac{\partial}{\partial x}(2x+1) + \frac{\partial}{\partial y}(x^2-6y) + \frac{\partial}{\partial z}(xy^2+3z)$

2 – 6 + 3 = –1 (–ve sink)

101.(d)

102.|F| = 40N

|D| = 3M
$\theta = 60^0$
W = |F| |D| cos θ
= 40 x 3 x $\frac{1}{2}$
D = 60J ans.

103.(d)
$V_p = 4$
$|4| = \begin{vmatrix} m & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}$
|4| = m (-1)–1 (–2) + 1 (3)
|4| = –m + 5
+4 = –m +5
M = 1, 9

104.(a)
|d| = 25 m
|F| = 180
$\theta = 45^0$
W = |F| |d| cos θ
= 25 x 180 x $\frac{1}{\sqrt{2}}$
$\frac{45000}{1,4}$
= 3.18 198 x 10^3J

105. (b) $\left.\begin{matrix} \lambda^2 i + j + k \\ i + \lambda^2 j + k \\ i + nj + \lambda^2 k \end{matrix}\right\}$ coplanar

$\begin{vmatrix} \lambda^2 & 1 & 1 \\ 1 & \lambda^2 & 1 \\ 1 & 1 & \lambda^2 \end{vmatrix} = 0$
$\lambda^2 (\lambda^4 - 1) - 1 (\lambda^2 - 1) + 1 (1 - \lambda^2) = 0$
$\lambda^6 - \lambda^2 - \lambda^2 + 1 + 1 - \lambda^2 = 0$
$\lambda^6 - 3\lambda^2 + 2 = 0$
$\lambda^6 - \lambda^2 - 2\lambda^2 + 2$
$\lambda^2(\lambda^4 - 1) - 2(\lambda^2 - 1)$
$\lambda^2(\lambda^2 - 1)$ $(\lambda^2 i)$-2$(\lambda^2 i)$=0
$(\lambda^2 - 1)$ $(\lambda^4 + \lambda^2 - 2) = 0$
$\lambda = \pm 1$
2 value of λ

1.